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1.0 Introduction

1.1 Dataset Description

The data to be used in this project is the World Happiness Report 2024. It has statistics of about 143 countries. The data set consists of such measures as the "Ladder score" (Happiness Score), Log GDP per capita, Social support, Healthy life expectancy, Freedom to make life choices, Generosity, and Perceptions of corruption.

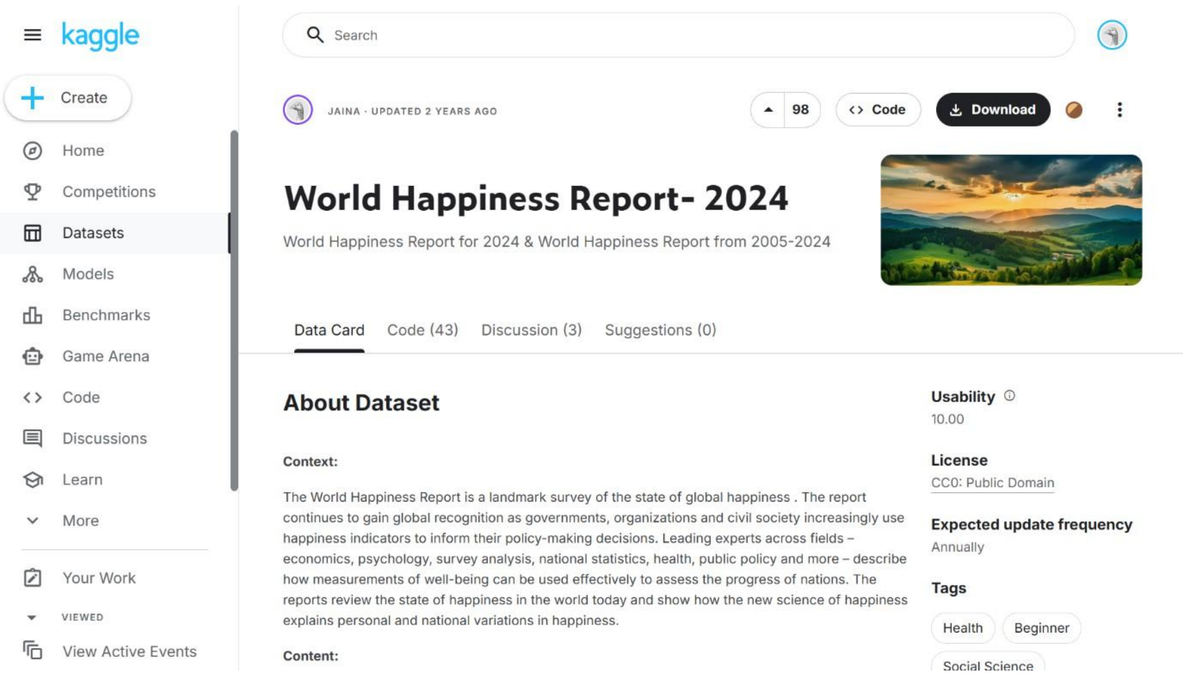
1.2 Project Objective

The purpose of this project is to examine the variables that result in an individual to achieve a higher national happiness score. Through the cleaning and visualization of the data, I will potentially figure out what is the relationship between economic/social metrics and the overall happiness of a population.

1.3 Analysis Questions

The project attempts to respond to the following questions

- What is the variation of happiness in various parts of the world?
- Does the happiness level of a nation relate significantly with the level of wealth (GDP) of the country?
- How does perceptions of corruption affect the happiness of a nation?
- What variable (GDP, Social Support, Life Expectancy, etc.), in particular, correlates best with the Happiness Score?



2.0 Data Preparation and Cleaning

Even though this dataset was relatively usable we organized it more to have a smoother coding experience.

2.1 Data Import and Inspection

The data was imported into R using the read.csv() function. The data was inspected using head(), str(), and summary().

```r
library(tidyverse)

# Loading the data
happiness_data <- read.csv("World-happiness-report-2024.csv")


# Data inspection
head(happiness_data)
str(happiness_data)
summary(happiness_data)
```

2.2 Data Cleaning

To clean the dataset with the help of the tidyverse library, the following steps were taken:

1. All of the originally used column names had their format modified to better convey their meaning. For example, Ladder.Score became score and Log.GDP.per.Capita became GDP.
2. Columns that provided no value for this analysis (i.e., Standard Error UpperWhisker and LowerWhisker), were removed from the data set, but the remaining columns (Country, Region, Score, GDP, Social support, Life expectancy, Freedom, Generosity, Corruption) were preserved so that all variables needed for analysis would be present.
3. Rows containing NA values were deleted using the drop_na() function so that the maintenance of statistical integrity through no missing data was assured.

```
# Data cleaning
clean_data <- happiness_data %>%
  # 1. Renaming the columns to simpler names
  rename(
    Country = Country.name,
    Region = Regional.indicator,
    Score = Ladder.score,
    GDP = Log.GDP.per.capita,
    Social_Support = Social.support,
    Life_Expectancy = Healthy.life.expectancy,
    Freedom = Freedom.to.make.life.choices,
    Corruption = Perceptions.of.corruption
  ) %>%

  # 2. Selecting only the columns we will use in the analysis
  select(Country, Region, Score, GDP, Social_Support, Life_Expectanc

  # 3. Droppping rows with missing values
  drop_na()
```

2.3 Creating User-Defined Functions

In order to analyze the data and satisfy the requirements of this project, I created three custom user-defined (R) functions that can be applied to the data set.

1) get_happiness_category(score) - This function takes a numeric score and assigns a "High," "Medium," or "Low" category of happiness to each country based on the score of the country in question. A new categorical column was created from this function that assists in the grouping of the data and its relationships visually through charts and graphs.

```
get_happiness_category <- function(score) {
  if (score >= 7) {
    return("High Happiness")
  } else if (score >= 5) {
    return("Medium Happiness")
  } else {
    return("Low Happiness")
  }
}
```

2) get_range(x) will take each column (containing numeric value(s)) in find_the_maximum_from_a_set() and compute the distance from the highest number to the

lowest number. By computing the difference between the highest and lowest numbers for the happiness scores (~6.02), the overall well-being variance across the world was identified.

```
get_range <- function(x) {
  if(is.numeric(x)) {
    return(max(x, na.rm = TRUE) - min(x, na.rm = TRUE))
  } else {
    return(NA)
  }
}
```

3) The normalize_column(x) function uses Min-Max scaling so all numeric values fall into a range of 0-1; it was used to scale the Score variable so that we can check for consistency within our data and also compare this variable to other variables that may have different units (for example, the GDP).

```
normalize_column <- function(x) {
  return ((x - min(x, na.rm = TRUE)) / (max(x, na.rm = TRUE) - min(x, na.rm = TRUE)))
}
```

3.0 Exploratory Data Analysis (EDA)

Regional Grouping: There was a particular analysis done in order to discover the average happiness score by region.

```
region_summary <- clean_data %>%
  group_by(Region) %>%
  summarise(Average_Score = mean(Score))
```

Correlation Analysis A correlation matrix was calculated in order to determine the best predictors of happiness:

```r
numeric_cols <- clean_data %>%
  select(Score, GDP, Social_Support, Life_Expectancy, Freedom, Generosity, Corruption)

cor_matrix <- cor(numeric_cols)

score_correlations <- cor_matrix[, "Score"]

score_correlations <- score_correlations[names(score_correlations) != "Score"]

ranking_df <- data.frame(
  Variable = names(score_correlations),
  Correlation = score_correlations
)
```

In accordance with the visual trends in the data, the variables that have the most positive associations with Happiness are GDP per Capita and Social Support.

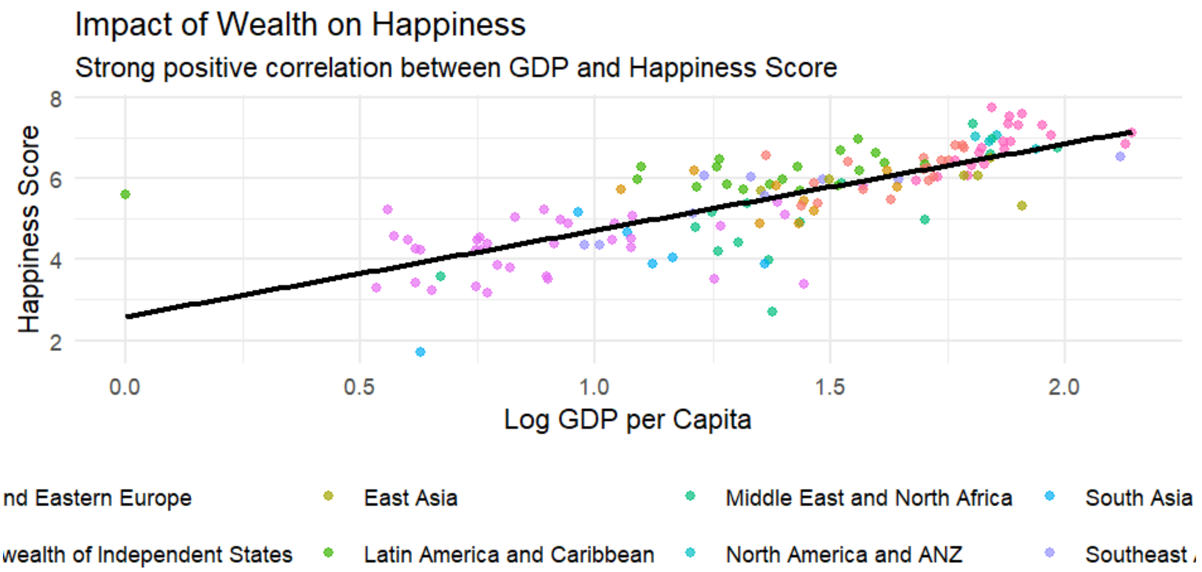
```
        Variable Correlation
  Social_Support   0.8135420
             GDP   0.7685037
 Life_Expectancy   0.7596594
         Freedom   0.6444511
```

4.0 Findings & Visualization

Now I'm responding to the questions that I had posed on the beginning of this report.

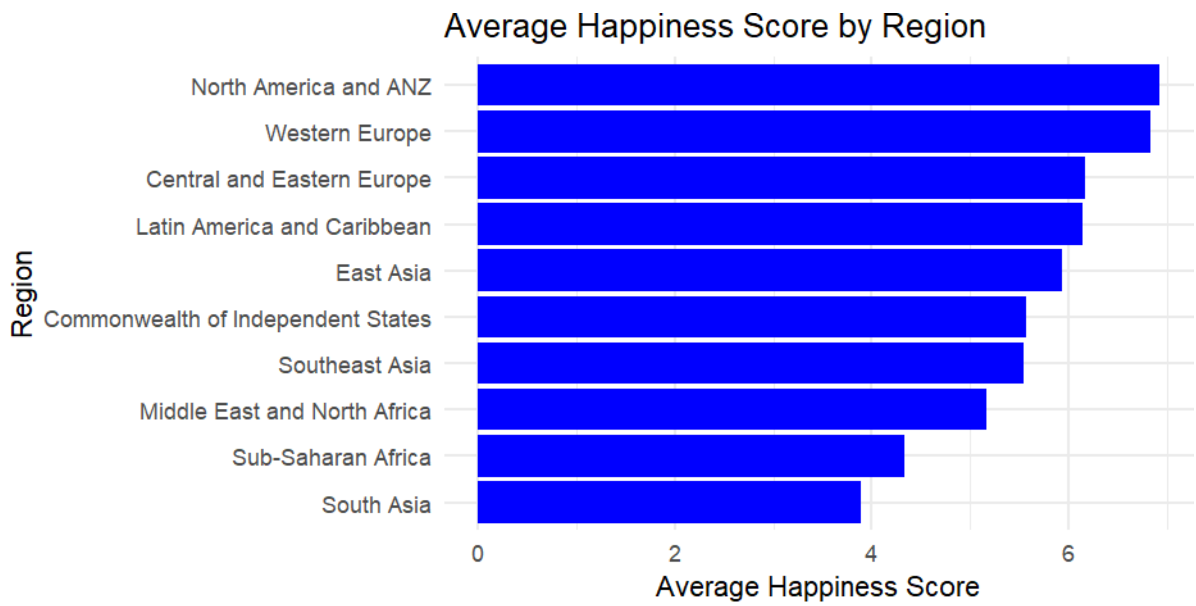
4.1 Wealth Matters

GDP per Capita and Happiness Scores have a clear strong positive correlation. The richer countries are happier.



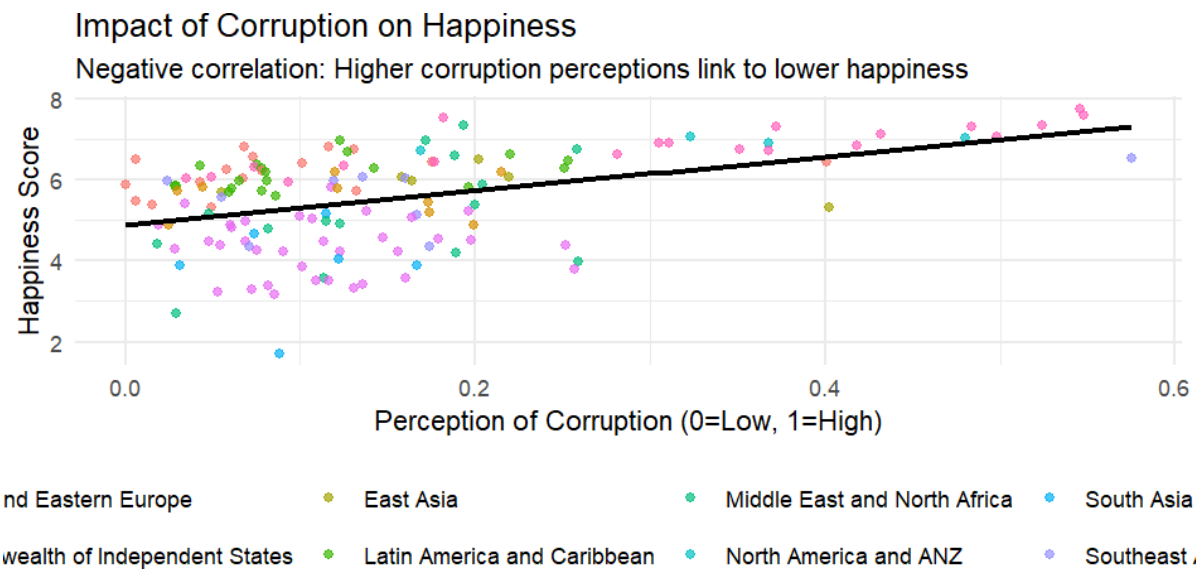
4.2 Regional Inequality

Happiness is not even distributed across the world. Western Europe is always rated higher whereas regions like South Asia and Sub-Saharan African are rated lower.



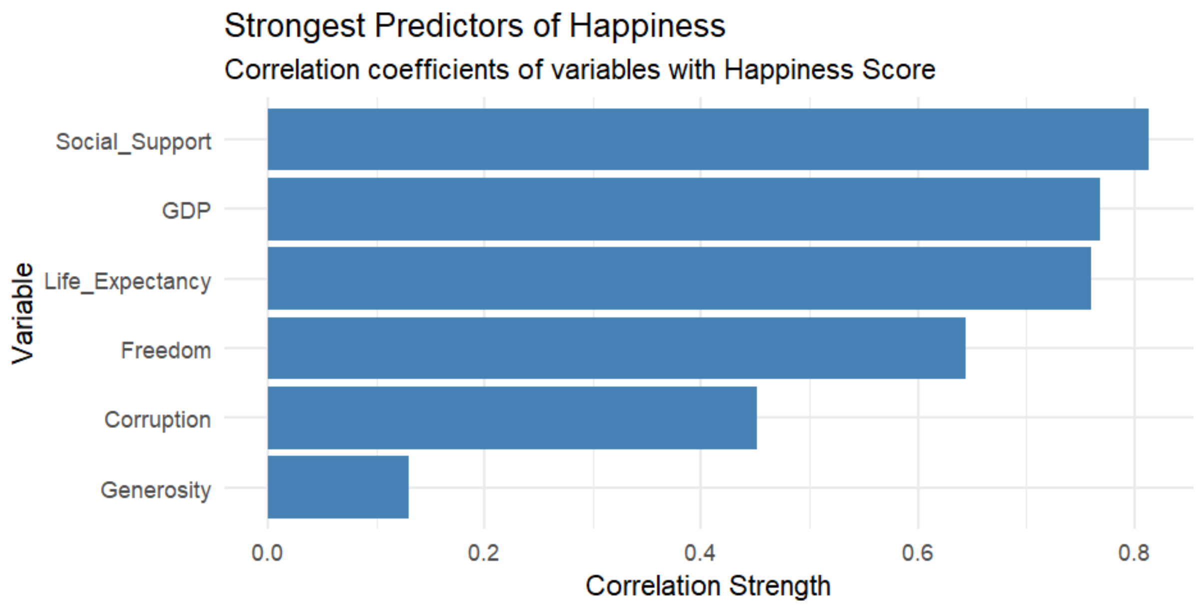
4.3 Corruption is a Bad Thing

The more countries are perceived to be corrupt, the lower their happiness scores are, which makes it clear that institutional distrust hurts well-being.



4.4 Social Support is Paramount

The correlation between the Happiness Score and Social Support is the strongest and positive, as shown in our correlation analysis. This implies that community and the availability of trusted friends or family to rely on during the bad times is the most important issue to national happiness as shown by this particular analysis to be even stronger than economic wealth.



5.0 References

[World Happiness Report 2024]